KARACA

SHIP WINDOWS & WIPERS

HEATED SHIP WINDOWS HEATED AREA DESIGN GUIDELINE – EN

Design principles for rectangular and trapezoidal bridge windows

1. Purpose and scope

The purpose of this document is to explain the design principles for determining the geometry of the heated area in heated ship windows used on the bridge and in navigation spaces. For all four-sided windows with rectangular, square or trapezoidal form, the sizing and positioning of the heated area is defined in a way that is compatible with ISO 3434 "Heated glass panes for ships' rectangular windows" and ISO 614 "Toughened safety glass panes for rectangular windows and side scuttles – Punch method of non-destructive strength testing".

The main objective of the guideline is to clarify why, especially for trapezoidal windows, making the heated area follow the complete outline of the glass is technically problematic, and to provide a common reference with the customer during the design phase.

2. Normative references

- The following standards are an integral part of this guideline:
- ISO 3434 – Ships and marine technology – Heated glass panes for ships' rectangular windows
- ISO 614 – Ships and marine technology – Toughened safety glass panes for rectangular windows and side scuttles – Punch method of non-destructive strength testing
- ISO 21005 – Ships and marine technology – Thermally toughened safety-glass panes for windows and side scuttles

3. Definitions and dimensions

The following definitions are used in heated glass drawings (see Figures 1–3):

- W1 / W2: Glass width.
- H1 / H2: Glass height.
- Heated area: Net area in which the heating element is active and heat output is defined in W/dm^2.
- Main clear vision area: Area that must be kept free of mist and ice, measured inwards from the edges by a certain distance.

It is not mandatory that the heated area follows the glass geometry one-to-one. Especially for trapezoidal windows, narrow triangular regions left on the long-corner side do not cause any non-conformity with the standards, as long as they remain outside the main clear vision area.

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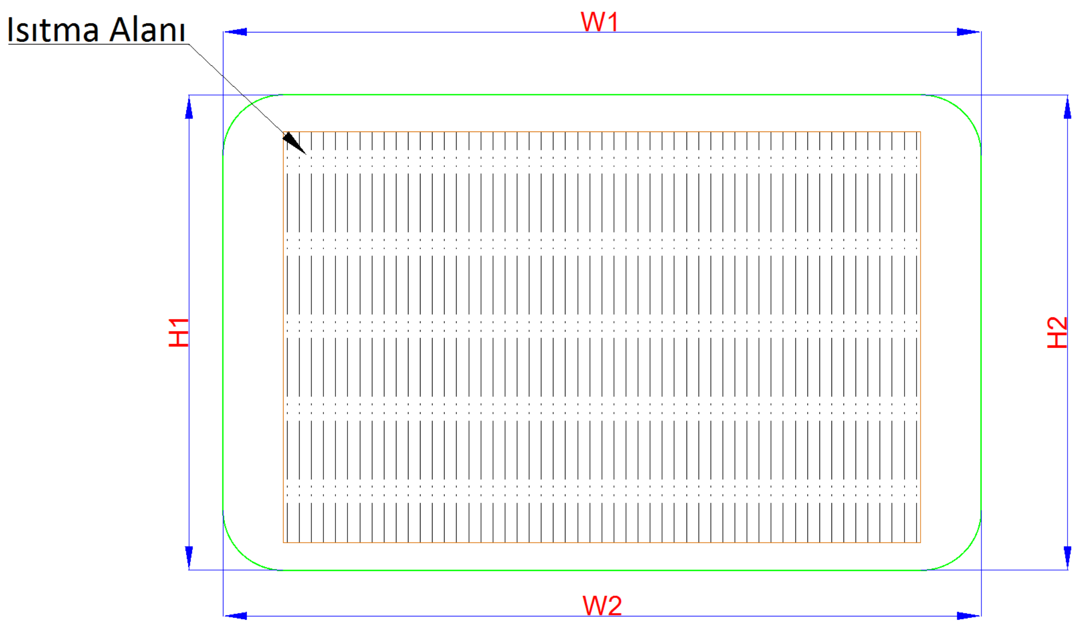
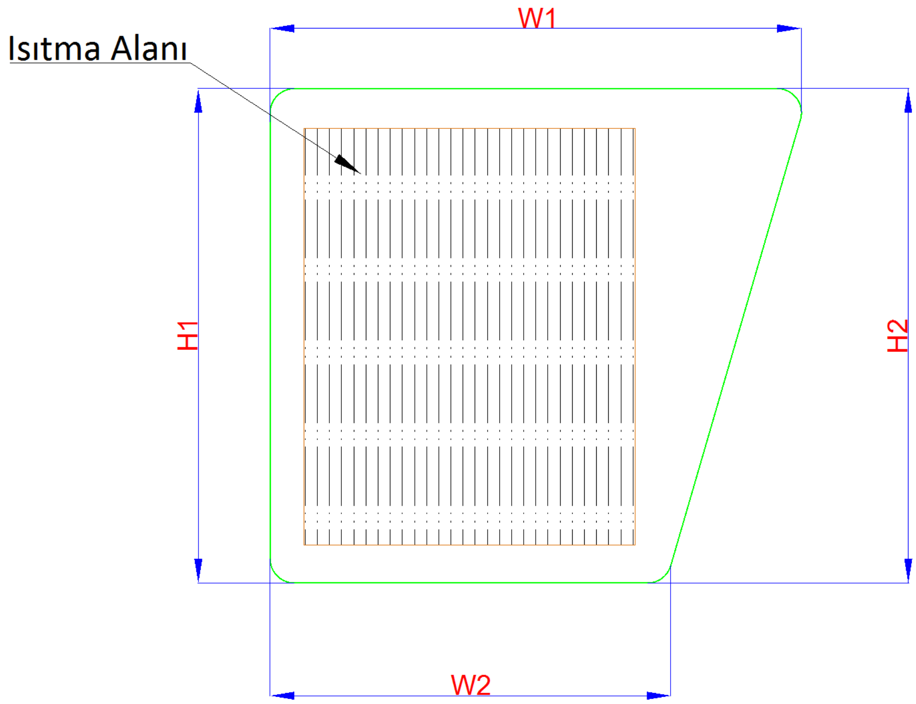
4. Heated area design principles

4.1 Rectangular windows – reference solution (Figure 1)



When the heated area is also rectangular in rectangular panes, the heating strips work between two parallel busbars with equal spacing and equal length. Since the electrical resistance of each strip is the same, the current and the heat output per unit area (W/dm^2) are practically homogeneous under a constant supply voltage. The power ranges given in ISO 3434 can therefore be safely achieved over the entire glass surface.

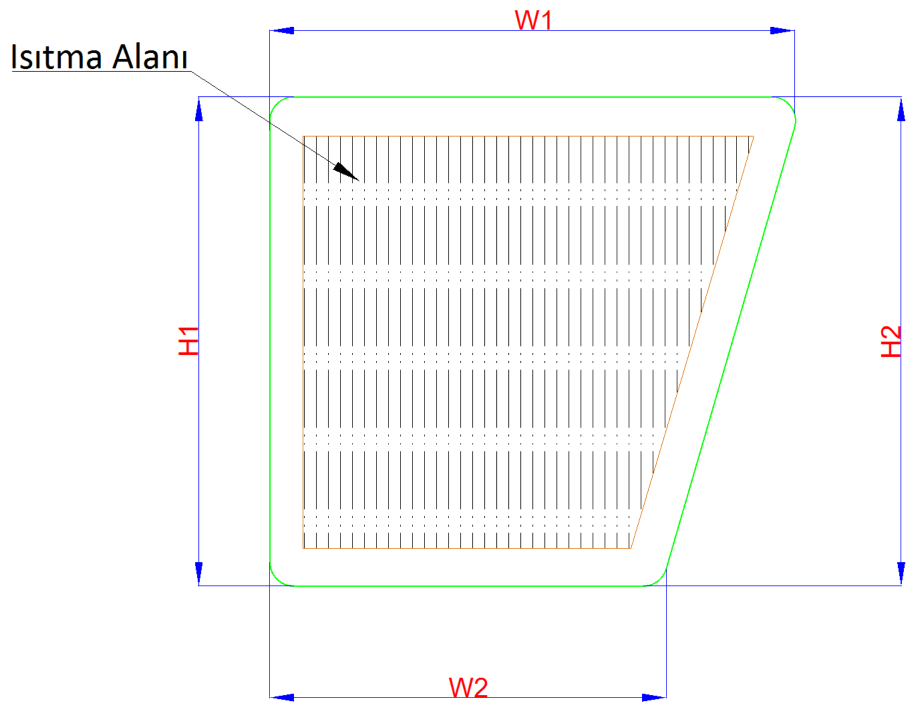
4.2 Recommended heated area for trapezoidal windows (Figure 2)



For trapezoidal windows, the recommended solution is to design the heated area as a rectangle placed along the short side of the glass. In this way, the heating strips remain equal in length and the same electrical model as for the rectangular pane is preserved.

With this arrangement, the main clear vision area remains completely inside the heated area. The narrow triangular area remaining towards the trapezoidal corner is usually a secondary region close to the frame; limited mist or ice formation in this area does not affect bridge visibility. From a design and certification point of view, this is the safe and preferred solution.

4.3 Full-area heating demands in trapezoidal windows (Figure 3 – why it is not recommended)



In some projects it is requested that, in a trapezoidal window, the heated area follows the glass geometry exactly and that there is no unheated region anywhere on the pane. Although this approach may look attractive in theory, it leads to serious practical problems:

1. Near the short side, the heating strips become shorter, so their resistance drops; therefore the current and W/dm² in this region increase, while they decrease on the long side. The heat distribution is no longer homogeneous.
2. Local high W/dm² can cause the glass surface temperature to approach around 50 °C despite the regulator setting, and leads to additional thermal stress within the lamination. This increases the risk of cracking and delamination.
3. To correct the non-uniformity, the manufacturer would have to change wire cross-section or spacing regionally, which significantly increases cost, production time and risk of manufacturing errors.
4. Because of the temperature difference between short- and long-side regions, differences in optical refraction and light scattering increase, especially during night navigation, and disturb the clear view along the wiper pattern.

For these reasons, designing the heated area to follow the full trapezoidal geometry of the glass is not technically recommended by our company. Standard solutions only cover the rectangular heated area geometries shown in Figures 1 and 2.

5. Selection of heating power according to climate zones

According to ISO 3434, heat output for heated panes is defined in three main power ranges. These ranges are selected with respect to the lowest outside design temperature:

- 7–9 W/dm^2 → regions where outside temperature is down to approx. −12 °C
- 12–15 W/dm^2 → regions where outside temperature is down to approx. −28 °C
- 17–21 W/dm^2 → regions where outside temperature is down to approx. −40 °C

In practice, these values can be associated with the typical climate zone in which the vessel operates. The following table summarizes typical guidance selections (requirements of the classification society and the owner's specification always take precedence):

Climate zone / Operating area	Design outside temperature	Recommended heat output range (W/dm^2)
Temperate / warm climate (Mediterranean, Aegean, milder parts of Black Sea etc.)	0 … −10 °C, short-term freezing	7–9 W/dm^2
Cold temperate climate (North Sea, large parts of North Atlantic and North Pacific etc.)	−10 … −25 °C, regular winter conditions	12–15 W/dm^2
Cold / sub-arctic climate (Baltic Sea, Scandinavian coasts, severe winter conditions)	−25 … −40 °C, risk of icing and significant snow load	17–21 W/dm^2
Polar regions and icebreakers (special projects)	−40 °C and below, heavy ice / sea ice	Above 21 W/dm^2, special design – to be agreed with manufacturer and class

ISO 3434 explicitly states that higher heat outputs may be required in polar regions and that, in such cases, the values must be evaluated on a project basis together with the glass manufacturer. Therefore, all solutions above 21 W/dm^2 are treated as special designs outside the standard ranges.

6. Electrical and thermal limits

- Heat output is calculated in W/dm^2 over the net heated area, not over the total glass size.
- The surface temperature of heated glass must be limited, by means of thermostat/regulator, so as not to exceed approx. 50 °C.
- The carrier glass must be thermally toughened safety glass in accordance with ISO 21005 and must pass the punch test according to ISO 614.
- Electrical supply must match the vessel's standard voltage system (e.g. 24 V DC, 115/230/440 V AC, etc.).

7. Conclusions on heated area

In trapezoidal ship windows, designing the heated area to follow the entire glass geometry in a trapezoidal form may give the first impression that the glass is "fully heated", but in practice it disturbs heat distribution, causing local overheating near the short side and insufficient heating near the long side. This increases the risk of deviation from the homogeneous W/dm^2 ranges defined in ISO 3434, and makes it harder to control surface temperature, thereby increasing stress and cracking potential.

For this reason, in bridge heated glass designs our company uses a rectangular heated area placed along the short side as the standard solution for trapezoidal windows, and does not recommend full trapezoidal heated areas for technical and safety reasons. Special requests must be treated as separate engineering work and require explicit approval from the manufacturer.

8. Selection of heating box

This section defines the selection of heating boxes used with heated glass and summarizes the application areas of manual and digital solutions. Selections must be evaluated together with the heated area and power values explained in the previous sections.

8.1 General principles

The primary function of the heating box is to control the power supplied to the heating element in the glass within a safe range according to glass type, climate zone and required surface temperature. The following criteria shall be considered when selecting the box:

- Glass type (single laminated only, combinations of insulated glass unit + lamination, etc.).
- Heat output range (e.g. 7–9 / 12–15 / 17–21 W/dm^2) and supply voltage (24 V DC, 115/230/440 V AC etc.).
- Control requirement: is a fixed switch-on/switch-off temperature sufficient, or are adjustable parameters and sensor feedback needed?
- Requirements of classification society and owner's specification (requirement for sensor-based / digital control etc.).

As a general rule, for simple, single-laminated glass and standard power ranges (especially 7–9 W/dm²), manual heating boxes are sufficient. For insulated glass units, multi-layer laminations and/or higher power ranges, sensor feedback and adjustable parameters are required, therefore a digital heating box is preferred.

8.2 Manual heating boxes (mechanical thermostat)

In manual heating boxes, control is provided by a mechanical thermostat in contact with the glass surface. The thermostat is not programmable; its factory setting typically switches on at approx. 25 °C and off at approx. 45 °C. The thermostat is mounted in direct contact with the glass pane or frame.

The box is equipped with an optional ON/OFF push-button. This button can be completely omitted on request, or relocated to another position such as a console panel / bridge panel. Inside the box, glass connection terminals, power input terminals and thermostat connections are standard; although box sizes vary, the electrical content is identical.

Main advantages of manual boxes:

- Simple and easy-to-understand internal wiring; easy installation and troubleshooting on board.
- No electronic PCB, therefore more tolerant to voltage fluctuations.
- Low spare part and maintenance cost.

Main limitations:

- Switch-on and switch-off temperatures are fixed; there is no possibility for user-adjustable parameters such as P1/P2.
- Only the outer surface temperature of the glass is sensed; the real temperature within the lamination is not monitored.
- For very cold climates, high W/dm² and insulated glass combinations, they may be insufficient in terms of safety and homogeneity.

Therefore, for single laminated bridge windows operating in standard climate conditions with 7–9 W/dm², manual heating boxes are considered the reference solution, unless the customer explicitly requests otherwise.

8.3 Digital heating box (programmable thermostat)

In the digital heating box, control is achieved by an electronic thermostat housed in an aluminium enclosure together with PT100 temperature sensors embedded in the glass lamination. As standard, two PT100 sensors are used; one is active and the other is a spare that can be connected in case of sensor failure. The sensor measures the lamination temperature between the glass plies and transmits it to the control unit.

The digital box parameters P1, P2 and P3 allow the operating characteristics to be adjusted:

- P1 – Control temperature (set point): target temperature of the glass. The maximum adjustable value is typically limited to 50 °C.
- P2 – Hysteresis: temperature band that prevents the device from switching too frequently. For example, with P2 = 2 °C, heating is switched on when temperature drops 2 °C below P1 and switched off when it reaches P1 again.
- P3 – Heater protection (duty-cycle setting): limits the time the heaters remain at full power by cycling the output periodically. Smaller values provide faster heating; higher values result in a smoother heating curve.

Main advantages of the digital heating box:

- More accurate control thanks to real temperature feedback from inside the lamination via PT100 sensors.
- Adjustable setpoint, hysteresis and heater protection functions for different glass types and climates.
- Quiet relay operation and use of the aluminium housing as natural heat sink.
- Safer solution for insulated glass and high W/dm^2 projects in line with class rules.

Therefore, for insulated and/or multi-layer laminated glass, as well as for front view windows operating in 12–15 / 17–21 W/dm^2 power ranges, the digital heating box is the standard solution of our company.

8.4 Solutions without local boxes / panel-based control

In some projects, a "clean bridge" appearance is requested and no boxes are allowed to be visible around the bridge windows. In such cases, solutions without local heating boxes may be used. Two basic options exist:

- All heating boxes are relocated into a separate electrical cabinet or control panel on board. Only cable terminations are taken out from the glass; the boxes are grouped inside the panel.
- Heating boxes are omitted completely and only heating and sensor cables from the glass are routed to the main ship switchboard. Switching and temperature control are performed entirely by the equipment in the ship's panel.

For such solutions, it is mandatory to follow Karaca's electrical diagrams and wiring instructions exactly. If cable cross-sections are incorrect, protective devices are missing, or connections are made wrongly, local overheating, cracking of the heated glass, burning of heating wires and damage to the ship's electrical system may occur. Solutions without boxes shall only be implemented by competent electrical personnel and with the approval of the responsible project engineer.

8.5 Heating box selection matrix

The table below summarizes typical combinations of heating box / control solutions. The actual selection must always be finalized on a project basis according to glass type, power range and class requirements.

Solution	Control type	Typical glass type	Standard selection / notes
Manual box – small / medium enclosures	Mechanical thermostat; fixed switch-on / switch-off temperatures; optional ON/OFF button.	Single laminated bridge windows, 7–9 W/dm², 220–230 V AC.	Reference solution when the customer does not specify special dimensions; box size is selected according to available installation space.
Manual box – large enclosure	Mechanical thermostat; larger terminal space and wiring volume.	Small bridges where several panes are supplied from the same group.	Preferred where maintenance and tidy cabling are important.
Digital heating box	Programmable thermostat; 2×PT100 sensors; P1 / P2 / P3 parameter settings.	Insulated or multi-layer laminated glass; 12–15 / 17–21 W/dm² power ranges.	Standard solution in class-controlled projects; parameters are recorded during commissioning.
No box – panel / cabinet solution	Control and protection entirely in the main ship switchboard; only cable exits from glass.	All glass types (assessed project-by-project).	Strict compliance with electrical diagrams and competent installation are mandatory; otherwise the risk of glass cracking and electrical damage increases.

8.6 Typical selection scenarios

The following examples can be used as quick guidance during offer and early design stages with the customer:

5. Standard bridge, temperate climate, single laminated glass: Heating power 7–9 W/dm^2; supply 220–230 V AC; reference solution is a manual heating box (small or medium enclosure).
6. Cold climate and insulated front view windows: Heating power 12–15 W/dm^2, or according to project specification; solution is a digital heating box with PT100 sensors embedded in the lamination.
7. High-safety applications – icebreakers / polar routes: Heating power 17–21 W/dm^2 and above; solution is a digital heating box or panel-based solution, to be evaluated jointly with the classification society on a project basis.

9. Appendices – Heating box dimensions and electrical diagrams

9.1 Heating box dimensions

The following technical drawings show typical outside dimensions of the standard heating box enclosures. All dimensions are given in millimetres. For current revisions, production drawings and product lists shall always take precedence.

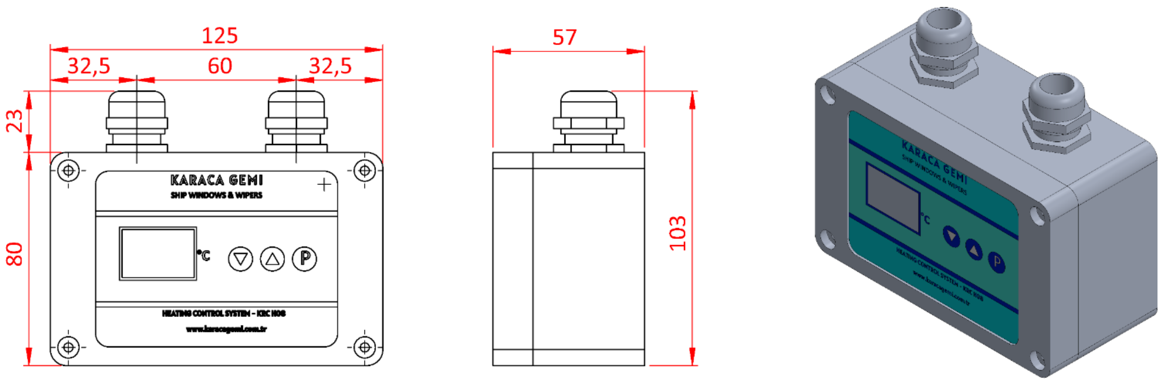


Figure – Typical outside dimensions of digital heating box (HB-100).

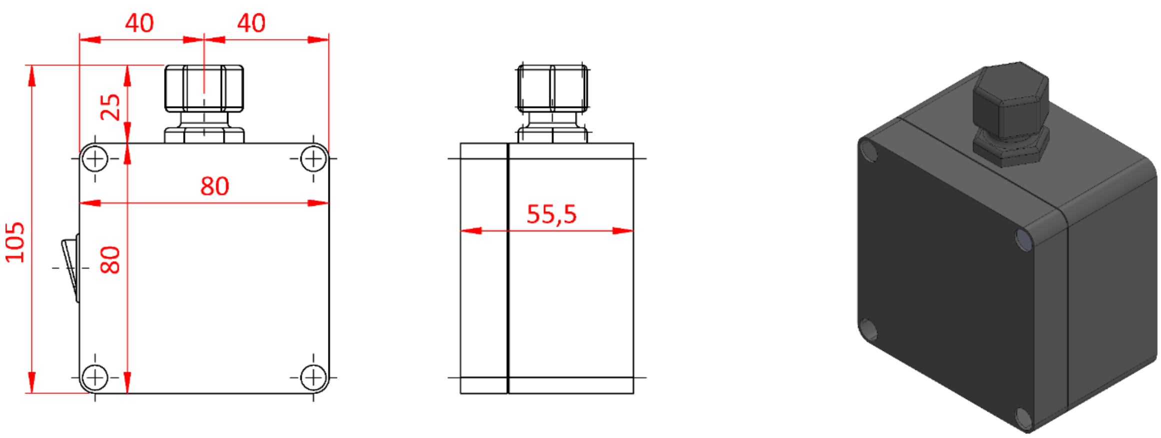


Figure – Typical outside dimensions of manual heating box – large enclosure (HB-200).

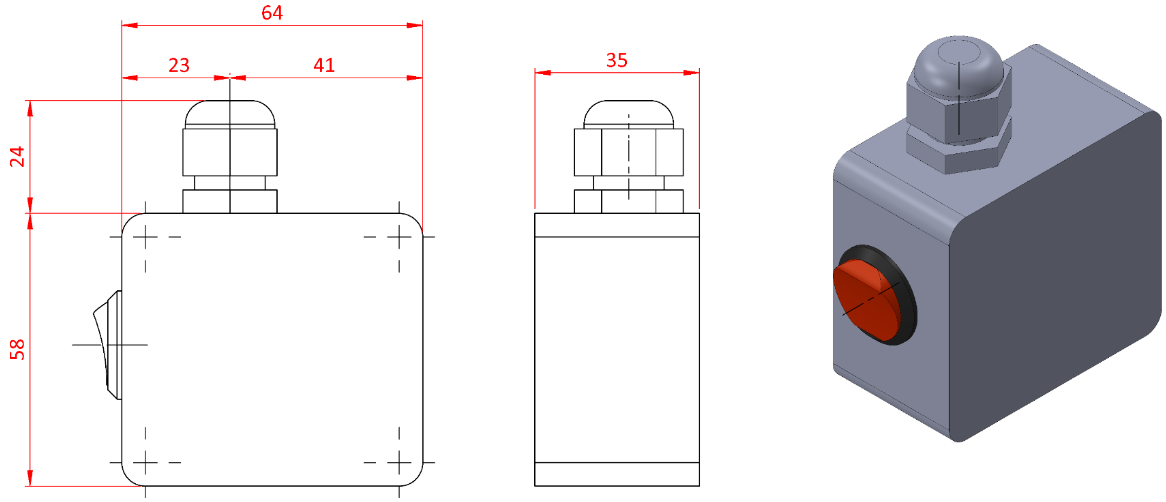


Figure – Typical outside dimensions of manual heating box – medium enclosure (HB-300).

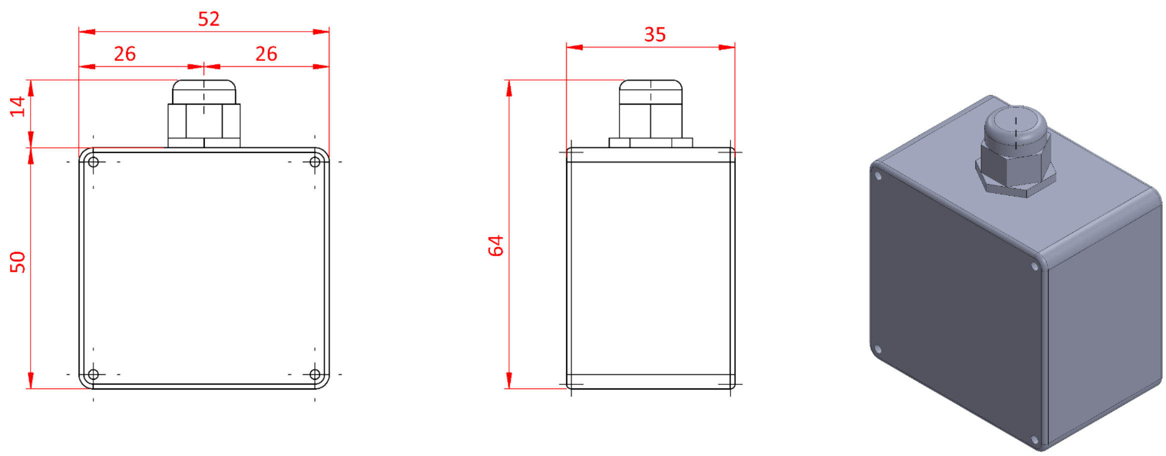


Figure – Typical outside dimensions of manual heating box – small enclosure (HB-400).

9.2 Electrical connection diagrams

These diagrams show typical connection methods of manual and digital heating boxes to heated glass circuits. In real projects, the final electrical drawings and class-approved diagrams shall always take precedence.

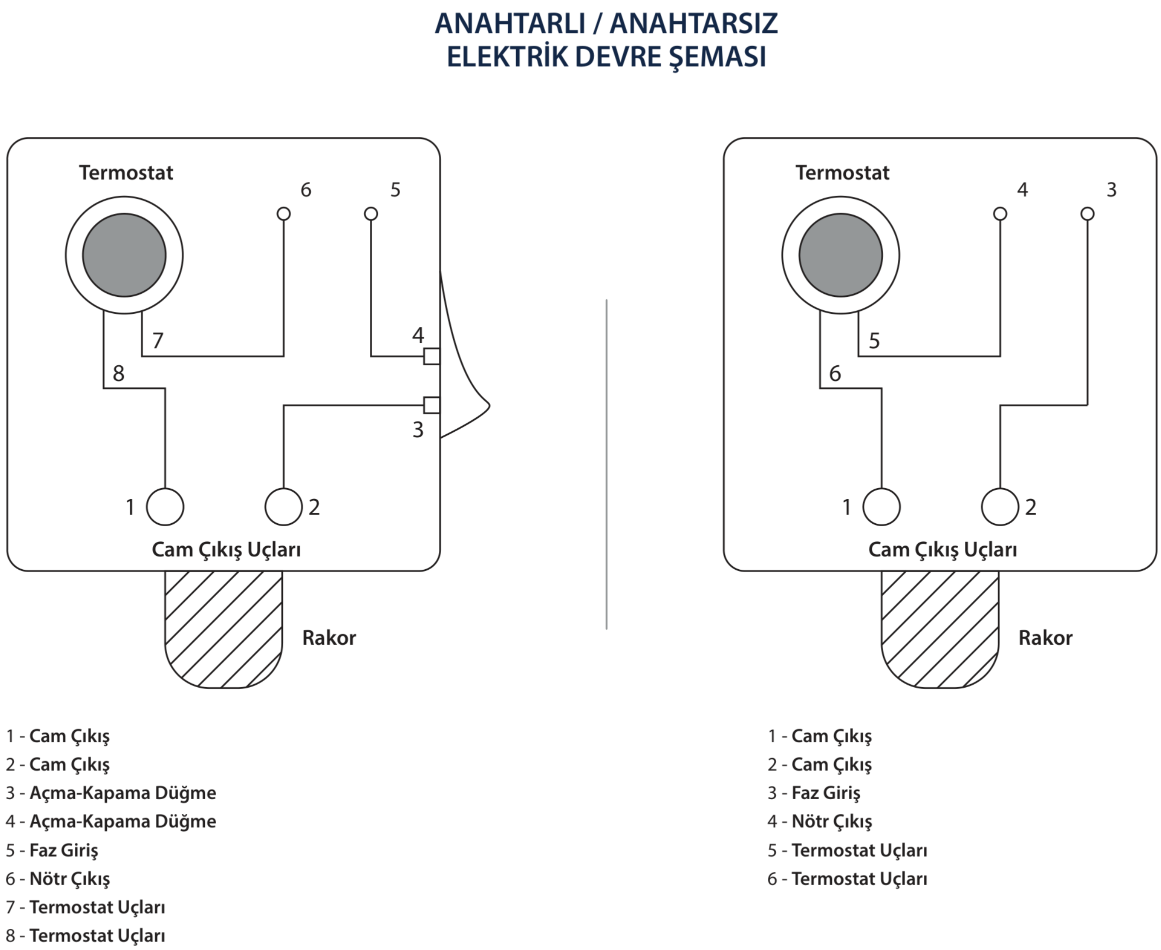


Figure – Typical electrical diagram for manual heating box (with / without switch).

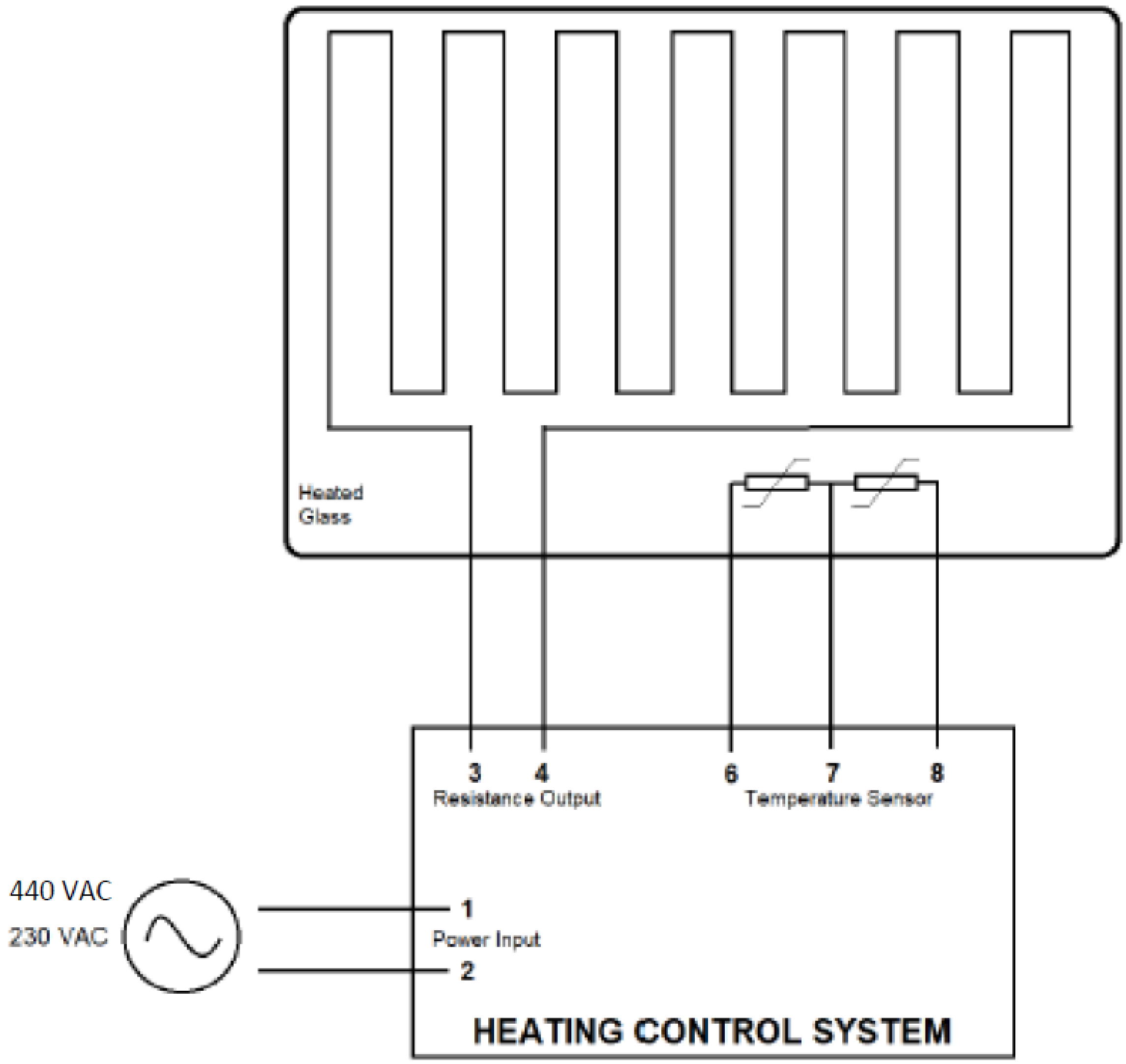


Figure – Electrical connection diagram for digital heating box and heated glass.

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This guideline has been prepared to be used as a technical reference document for heated glass designs of KARACA SHIP WINDOWS. For project-specific requirements and class-approved solutions, please contact our company.